KKK-T-654D
May 20, 1985
SUPERSEDING
KKK-T-654C
July 2, 1976

FEDERAL SPECIFICATION

TRUCKS, DOLLY (ROLLER)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for dolly trucks of all-steel frame construction with steel or aluminum rollers.

1.2 Classification.

1.2.1 Types and sizes.

Type I - Stevedore, 4500 pound capacity.
Size 1 - 24- by 72-inch frame.
Type II - Warehouse, 4000 pound capacity.
Size 1 - 30- by 30-inch frame.
Size 2 - 42- by 42-inch frame.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

QQ-S-781 - Strapping, Steel, and Seals.

FEDERAL STANDARDS

FED-STD-123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The index, which includes cumulative bimonthly

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supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Description from established distribution points in their agencies.)

MILITARY SPECIFICATIONS

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

MILITARY STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar metals.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations

41 CFR 1-1.25	- Recovered Materials.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society for Testing and Materials (ASTM)

A 36	- Structural Steel.
A 519	- Seamless Carbon and Alloy Mechanical Tubing.
B 210	- Aluminum-Alloy Drawn Seamless Tubes.
B 483	- Aluminum-Alloy Drawn Tubes for General Purpose Applications.
D 3951	- Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. Roller-type dolly trucks (hereinafter called "trucks") shall consist of a heavy steel frame with rows of steel or aluminum rollers at each side and below the carrying surface of the frame as shown in figures 1 and 2. The truck shall be the manufacturer's current commercial product.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of trucks, from the approved first article during production will be subjected to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Frame assembly.

3.3.1.1 Type I. The outer frame and lateral reinforcing members of the type I truck shall be fabricated from structural steel angles or formed angles. The center roller support plate and towing rings shall be fabricated from steel. The steel components shall conform to ASTM A 36 and the frame assembly shall be all-welded construction.

3.3.1.2 Type II. The outer frame and longitudinal frame stiffener and roller assembly stiffener of the type II truck shall be fabricated from structural steel angles, formed angles, or channels conforming to ASTM A 36, and shall be of all-welded construction.

3.3.2 Rollers. The rollers shall be fabricated from seamless commercial steel tubing (for types I and II trucks) conforming to ASTM A 519, grade 1015-1025, or high tensile aluminum alloy tubing (for type II trucks) conforming to ASTM B 210 or B 483, Alloy 6061-T6.

3.3.3 Material deterioration and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

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3.3.4 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.5 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.6 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 (41 CFR 1-1.25) to the maximum extent practical.

3.4 Construction. Construction of type I and type II trucks shall be in accordance with figures 1 and 2, respectively, and as specified herein. The rated capacity for type I and type II trucks shall be 4500 pounds and 4000 pounds, respectively.

3.4.1 Roller assembly. The roller assembly shall be equipped at each end with heavy-duty commercial-type ball bearings. The outer race of the ball bearing shall be assembled to the roller to roll with the roller. The inner race of the ball bearing shall be assembled on the roller shaft and shall be a pressed fit. The roller assembly with shaft shall be assembled to the frame assembly in such a manner as to prevent any lateral movement of the roller assembly, and shall be removable for replacement.

3.4.1.1 Ball bearings. Bearing load capacity shall withstand the maximum static load (see 3.5.1) when tested as specified in 4.5.2.1. The bearings shall be grease packed and shall retain grease and exclude foreign matter. Sealed-type bearings may be used.

3.5 Performance.

3.5.1 Static load. The trucks shall withstand a static load equal to 200 percent of rated capacity without any permanent deformation, weld failure, distortion, or any other damage to any part of the truck when tested as specified in 4.5.2.1.

3.5.2 Rolling resistance. The force required to tow the truck shall not exceed 2 percent of the gross weight. The gross weight shall include the weight of the truck and 100 percent of rated capacity.

3.6 Identification marking. Each truck shall be identified with an identification plate containing the following information: Nomenclature, contract number, national stock number (NSN), and manufacturer. The plate shall conform to MIL-P-514, type III, composition C (GG-5-455) type I, grade A, class D unless otherwise specified. The identifying data shall be stamped in letters in accordance with MIL-P-514. The plate shall be permanently attached to the truck.

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3.7 Treatment and painting. The parts of the truck normally painted shall be cleaned, treated and painted in accordance with the contractor's commercial practice.

3.8 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.9 Workmanship. All parts, components, and assemblies of the truck including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges or corners are required. Any defects that affect serviceability or appearance shall be rejected.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

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4.4.1 Examination. Each truck shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The trucks shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Dimensions not as specified.
106. Construction not as specified.
107. Ball bearings not as specified.
108. Lubrication not as specified.
109. Workmanship not as specified.
110. Roller assembly not as specified.
111. Type I and type II frame assembly not as specified.
112. Rollers not as specified.

Minor

201. Identification marking not as specified.
202. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Static load. The truck shall be placed on a level concrete surface and shall be loaded uniformly with a load equal to 200 percent of rated capacity, i.e., type I - 9000 pounds; type II - 8000 pounds. The truck shall remain under load for a period of 1 hour after which the load shall be removed and the truck shall be examined. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.2 Rolling resistance. The truck under test shall be weighed and then placed on a smooth level concrete roadway. The truck shall be loaded with a uniform load of 4500 pounds for type I and 4000 pounds for type II. The truck shall then be coupled through a tension dynamometer to a suitable towing device. The towing device shall accelerate the loaded truck to a speed of 2 miles per hour and maintain that speed for a minimum of 100 feet. During this 100-foot interval, the maximum gage reading on the dynamometer shall be recorded.

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Repeat the test towing the truck in the opposite direction. The average of the two dynomometer readings shall be considered the rolling resistance. Nonconformance to 3.5.2 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

113.	Trucks of unlike description secured together for level A.
114.	Strapping not as specified for level A.
115.	Strapping not applied as specified for level A.
116.	Commercial packing not in accordance with referenced document.
117.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Two trucks of like description shall be secured together by placing one truck upon the other back to back. Securing shall be by the use of three 0.035 x 3/4 inch flat steel straps conforming to QQ-S-781, class 1, type I or IV. Unless otherwise specified (see 6.2), strapping shall be finish A. Two straps shall be placed girthwise, approximately one-sixth the distance in from each end, and the third strap placed lengthwise approximately in the center.

5.1.2 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract, all containers shall be marked in accordance with FED-STD-123.

5.2.2 Military. In addition to any special markings required by the contract, marking for level A military shipments shall be in accordance with MIL-STD-129.

5.2.3 Commercial. Commercial packs shall be marked in accordance with ASTM D 3951.

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6. NOTES

6.1 Intended use. Trucks covered by this specification are intended for use in moving palletized and other loads over level hard surfaces for short distances. Type I trucks are primarily used for stevedore operations requiring frequent turning and power towing. Type II trucks are primarily used for moving loads in a straight line by manual means during loading and unloading of highway trucks, trailers, and boxcars from warehouse loading platforms.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type and size required (see 1.2.1).
c. Time frame required for submission of first article (see 3.2).
d. Degree of packing required (see 5.1).
e. When strapping shall be other than as specified (see 5.1.1).

6.3 First article. When a first article inspection is required, the items should be preproduction units. The first article should consist of one of each type and size. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents' first article

6.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

MILITARY INTEREST:

Custodians

Army - ME
Navy - SA
Air Force - 99

Review activities

Army - AT
Air Force - 84
DLA - GS

User activity

Navy - OS

CIVIL AGENCY COORDINATING ACTIVITIES:

DC GOVT-DCG
GSA-FSS
HHS-FEC
JUSTICE-FPI
VA-OSS

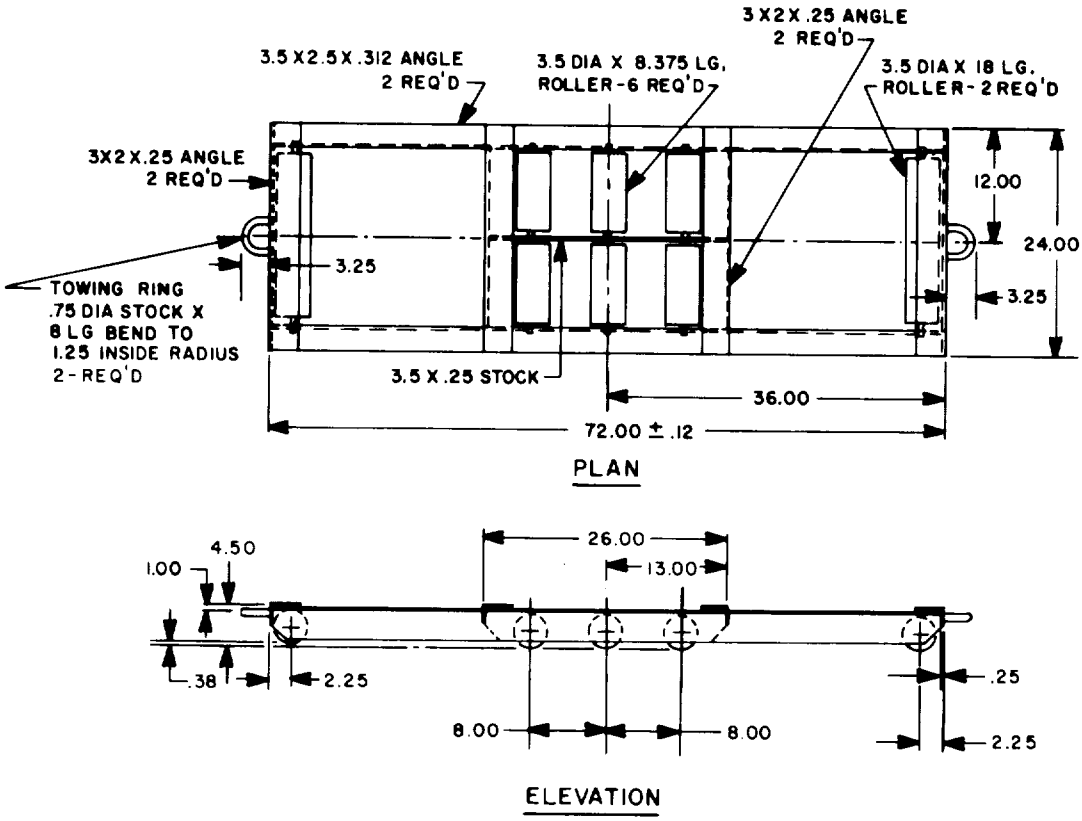
Preparing activity:

Army - ME

Project No. 3920-0160

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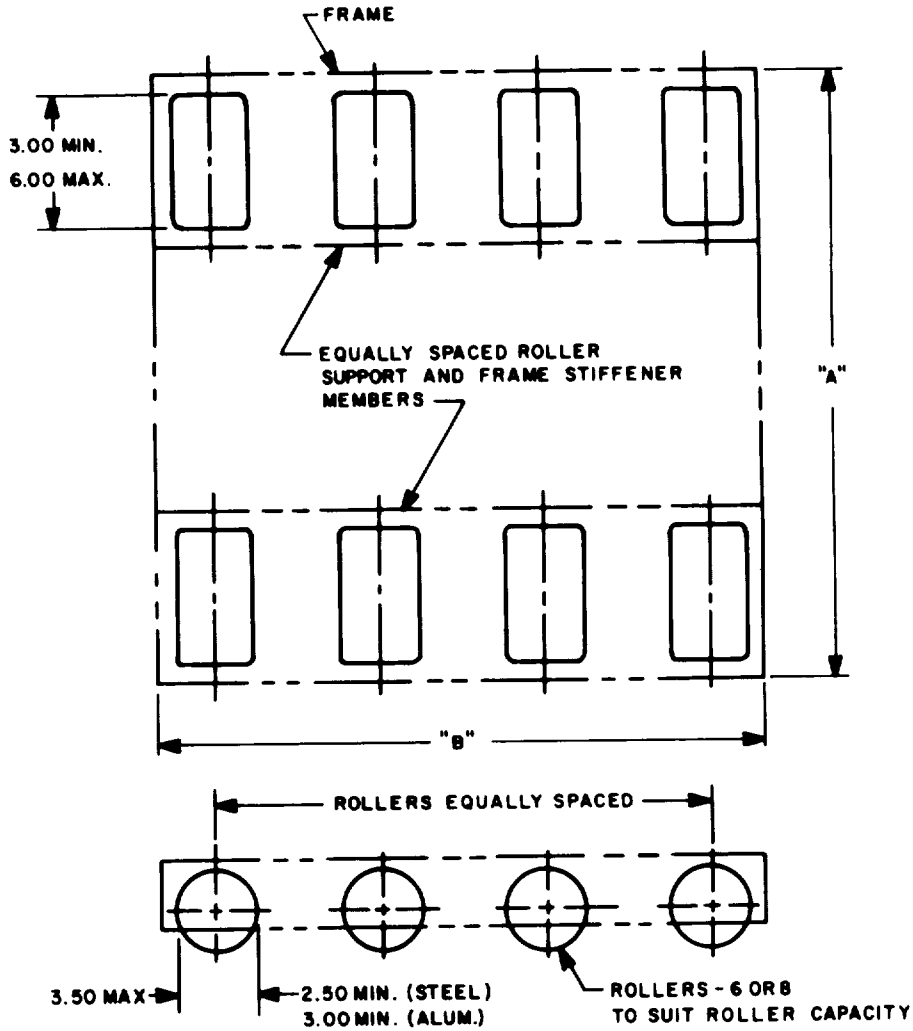
NOTES:

1. ALL WELDED CONSTRUCTION.
2. ROLLERS - TUBING, SEAMLESS, STEEL 3.5 O.D. X .312 (MIN.) WALL THICKNESS.
3. ROLLER AXLES - HEXAGONAL, COLD FINISHED STEEL, 1.062 ACROSS FLATS (MIN.). OPTION - ROUND, COLD FINISHED STEEL, 1.188 DIA. (MIN.).
4. ALL DIMENSIONS ARE IN INCHES.
5. UNSPECIFIED TOLERANCE ± .06.

FIGURE 1. Truck, dolly (roller), type I, 4,500 pounds capacity.

X-1721B

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SIZE	DIM "A"	DIM "B"
1	30.00	30.00
2	42.00	42.00

NOTES:
1. ALL DIMENSIONS ARE IN INCHES
2. TOLERANCE ±.06

FIGURE 2. TRUCK, DOLLY (ROLLER), TYPE II, 4,000 POUNDS CAPACITY
X-1722A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
KKK-T-654D	Trucks, Dolly (Roller)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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